

Scientific Foundations

The psychology and neuroscience sources that shaped Cortex - and the engineering decisions drawn from them.

25	8	1
PSYCHOLOGY + NEUROSCIENCE SOURCES	DESIGN PRINCIPLES	ENGINEERING INSPIRATION



**Psychology supplies hypotheses.
Benchmarks decide.**

Prepared for the Cortex Memory open-source project
Research map compiled July 2026

The thesis

Cortex is not a simulation of the human brain. It is an agent-memory system whose constraints were informed by well-established findings about attention, retrieval, association, consolidation, forgetting, and revision.

1 Memory is plural

Separate recent experiences, durable facts, and learned procedures instead of treating every record as the same kind of memory.

2 Attention is bounded

Retrieve a small, high-value working set. Sometimes the right amount of recalled memory is zero.

3 Access is adaptive

Recency, frequency, spacing, relevance, and successful use affect accessibility - but never determine truth.

4 Association is useful

Traverse a bounded graph of meaningful relationships rather than relying only on semantic similarity.

5 Consolidation is offline

Move expensive replay, summarization, contradiction checks, and graph repair outside normal inference.

6 Forgetting can help

Cool or archive stale, redundant, and low-value material to reduce interference and retrieval cost.

7 Recall can revise

Version corrected memories, preserve evidence, and rebuild dependent summaries instead of silently overwriting history.

8 Outcomes teach utility

A memory or procedure should strengthen when it helps and weaken when it causes retries, corrections, or failure.

The responsible claim

These studies generate design hypotheses. Cortex still has to earn its value through measured improvements in retrieval quality, task success, token use, latency, and pruning regret.

Memory architecture

Why Cortex separates kinds of memory, treats accessibility independently from storage, and keeps fast experience capture distinct from slow knowledge formation.

1

Human Memory

NCBI Bookshelf. Overview of interacting human memory systems.

Cortex influence: A multi-store architecture rather than one undifferentiated database. [Open source](#)

2

Squire - Memory systems of the brain (2004)

Neurobiology of Learning and Memory. PMID 15464402.

Cortex influence: Explicit episodic, semantic, and procedural roles in the memory model. [Open source](#)

3

Squire et al. - Memory Consolidation (2015)

Cold Spring Harbor Perspectives in Biology. PMCID PMC4526749.

Cortex influence: A distinction between recent experiences and consolidated long-term knowledge. [Open source](#)

4

Tonegawa et al. - Memory engram storage and retrieval (2015)

Current Opinion in Neurobiology. PMID 26335640.

Cortex influence: Storage, accessibility, confidence, and current retrieval strength are tracked separately. [Open source](#)

5

McClelland, McNaughton & O'Reilly - Complementary Learning Systems (1995)

Psychological Review. DOI 10.1037/0033-295X.102.3.419.

Cortex influence: Fast episodic capture paired with slower semantic and procedural aggregation. [Open source](#)

ENGINEERING TRANSLATION

Episodes preserve what happened. Semantic memories preserve what Cortex currently believes. Procedures preserve how to act. Consolidation can connect them without erasing the original evidence.

Attention, retrieval, and association

The sources behind Cortex's token-bounded attention gate, adaptive accessibility score, associative graph, and outcome-aware reinforcement.

6

Baddeley - The episodic buffer (2000)

Trends in Cognitive Sciences. PMID 11058819.

Cortex influence: Bounded context assembly that integrates useful information for the current task. [Open source](#)

7

Cowan - The magical number 4 in short-term memory (2001)

Behavioral and Brain Sciences. PMID 11515286.

Cortex influence: Strict top-k retrieval and token budgets; the whole memory store never enters context. [Open source](#)

8

Anderson & Schooler - Reflections of the Environment in Memory (1991)

Psychological Science. DOI 10.1111/j.1467-9280.1991.tb00174.x.

Cortex influence: Recency and frequency contribute to accessibility, while remaining separate from truth. [Open source](#)

9

Collins & Loftus - Spreading-Activation Theory (1975)

Psychological Review. DOI 10.1037/0033-295X.82.6.407.

Cortex influence: Explicit associations and bounded graph activation supplement vector retrieval. [Open source](#)

10

Roediger & Karpicke - Test-enhanced learning (2006)

Psychological Science. PMID 16507066.

Cortex influence: Successful use reinforces a memory; being returned by search does not automatically do so. [Open source](#)

11

Cepeda et al. - Distributed practice (2006)

Psychological Bulletin. Meta-analysis; PMID 16719566.

Cortex influence: Spaced, distinct successful uses count more than repeated calls in one burst. [Open source](#)

QUERY	CANDIDATES	ATTENTION GATE	OUTCOME
What is the agent trying to do?	Semantic + lexical + graph retrieval	Small token-bounded working set	Helpful, unused, harmful, or corrected

Offline consolidation: Sleep

Cortex moves high-level reflection outside the latency-sensitive inference path. Sleep selectively replays prior experience, repairs the graph, and proposes durable knowledge.

12

Klinzing, Niethard & Born - Systems consolidation during sleep (2019)

Nature Neuroscience. PMID 31451802.

Cortex influence: The broad scientific frame for offline consolidation. [Open source](#)

13

Wilson & McNaughton - Reactivation during sleep (1994)

Science. DOI 10.1126/science.8036517.

Cortex influence: Selective replay of prior episodes during offline processing. [Open source](#)

14

Rudoy et al. - Strengthening memories through sleep reactivation (2009)

Science. DOI 10.1126/science.1179013.

Cortex influence: Replay can target particular memories instead of processing everything equally. [Open source](#)

15

Girardeau et al. - Suppression of hippocampal ripples impairs memory (2009)

Nature Neuroscience. DOI 10.1038/nn.2384.

Cortex influence: Replay-like offline activity is treated as functionally meaningful, not cosmetic maintenance. [Open source](#)

16

Ellenbogen et al. - Relational memory requires time and sleep (2007)

PNAS. DOI 10.1073/pnas.0700094104.

Cortex influence: Cross-session relationship discovery and connection proposals. [Open source](#)

Replay

Prioritize surprising, useful, unresolved, or weakly consolidated episodes.

Reorganize

Extract summaries, connect recurring entities, and flag contradictions.

Repair

Rebuild stale derivations, cool weak edges, and record every mutation.

Adaptive forgetting and pruning

Forgetting is treated as resource management and interference control. Cortex weakens or archives cautiously, with reversibility and measurable regret.

17

Langille - Remembering to Forget (2019)

Frontiers in Cellular Neuroscience. PMID 30930746.

Cortex influence: Sleep can support both consolidation and adaptive forgetting, motivating selective retention rather than blanket strengthening. [Open source](#)

18

de Vivo et al. - Synaptic scaling across wake and sleep (2017)

Science. DOI 10.1126/science.aah5982.

Cortex influence: Gradual offline downscaling of weak associations. [Open source](#)

19

Diering et al. - Homeostatic scaling-down during sleep (2017)

Science. DOI 10.1126/science.aai8355.

Cortex influence: Bounded weakening rather than indiscriminate deletion. [Open source](#)

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Norimoto et al. - Hippocampal ripples down-regulate synapses (2018)

Science. DOI 10.1126/science.aao0702.

Cortex influence: Replay and selective weakening can coexist in one maintenance cycle. [Open source](#)

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Li et al. - REM sleep selectively prunes and maintains synapses (2017)

Nature Neuroscience. DOI 10.1038/nn.4479.

Cortex influence: Some connections cool while important or recently useful ones remain protected. [Open source](#)

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Yang et al. - Sleep promotes branch-specific spine formation (2014)

Science. DOI 10.1126/science.1249098.

Cortex influence: Offline work also creates and stabilizes connections; it does not only remove them. [Open source](#)

ACTIVE	COOLING	COLD	ARCHIVED
Frequently useful or recently validated	Declining access or uncertain utility	Excluded from ordinary retrieval	Retained for audit and possible recovery

Revision, self-healing, and outcomes

Cortex preserves evidence while allowing beliefs and procedures to change. Retrieval success is judged by contribution to the task, including tool-use results.

23

Nader & Hardt - The case for reconsolidation (2009)

Nature Reviews Neuroscience. DOI 10.1038/nrn2590.

Cortex influence: Retrieved memories can be revised; Cortex versions corrections rather than silently overwriting history. [Open source](#)

24

Memory Reconsolidation

NCBI Bookshelf overview.

Cortex influence: Contradictions, supersession, dependency invalidation, and regeneration of derived summaries. [Open source](#)

25

Schultz - Dopamine reward prediction error coding (2016)

Current Opinion in Neurobiology. PMID 27069377.

Cortex influence: A loose inspiration for learning from helpful, harmful, corrected, or surprising outcomes - not a dopamine simulation. [Open source](#)

What this means for tool calling

OBSERVED SIGNAL	MEMORY UPDATE	EXPECTED EFFECT
Tool and arguments succeed	Strengthen the procedure for similar task states	Fewer exploratory calls
Call fails, then a correction works	Link failure context to the corrected procedure	Avoid repeated mistakes
Retrieved memory is ignored	Do not reward it merely for appearing	Cleaner ranking
Procedure repeatedly wastes tokens	Lower utility and retain the failure evidence	More efficient planning

Self-healing does not mean self-authorizing.

Cortex records provenance, versions mutations, preserves older evidence, and places confidence limits on consolidation. Offline maintenance should remain inspectable, reversible, and benchmarked.

From finding to measurable system

The research is useful only when each analogy becomes a concrete mechanism with a falsifiable engineering claim.

RESEARCH PRINCIPLE	CORTEX MECHANISM	MEASURABLE CLAIM
Multiple memory systems	Episodes + semantic knowledge + procedures	Task accuracy by memory type
Bounded working memory	Top-k and token-budgeted attention gate	Context tokens and retrieval latency
Adaptive accessibility	Recency, frequency, spacing, relevance, utility	Recall, precision, and stale-hit rate
Spreading activation	Bounded traversal of explicit memory edges	Multi-hop recall and graph cost
Retrieval practice	Reinforce only successful downstream use	Future hit rate after confirmed use
Sleep consolidation	Offline replay, summarization, repair, and linking	Online latency and post-sleep quality
Adaptive forgetting	Cooling, cold storage, archive, reactivation	Store size, interference, pruning regret
Reconsolidation	Versioned corrections and dirty-dependency rebuilds	Contradiction recovery and auditability
Outcome learning	Helpful, unused, harmful, corrected result signals	Tool success, retries, and tokens per task

Evaluation rule

Compare Cortex against the same agent and model using a default memory baseline. Report task success, retrieval precision/recall, context tokens, end-to-end latency, tool retries, maintenance cost, and false-pruning recovery. Avoid attributing model quality changes to memory architecture.

Complete linked bibliography

Primary and authoritative landing pages are provided below. The final entry is an adjacent continual-learning paper, not a psychology or neuroscience source.

- 1 **Human Memory**
NCBI Bookshelf [Open source](#)

- 2 **Squire. Memory systems of the brain: A brief history and current perspective**
Neurobiology of Learning and Memory, 2004. PMID 15464402 [Open source](#)

- 3 **Squire et al. Memory Consolidation**
Cold Spring Harbor Perspectives in Biology, 2015. PMCID PMC4526749 [Open source](#)

- 4 **Tonegawa et al. Memory engram storage and retrieval**
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- 6 **Baddeley. The episodic buffer: a new component of working memory?**
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Psychological Review, 1975. DOI 10.1037/0033-295X.82.6.407 [Open source](#)

- 10 **Roediger & Karpicke. Test-enhanced learning**
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- 11 **Cepeda et al. Distributed practice in verbal recall tasks**
Psychological Bulletin, 2006. PMID 16719566 [Open source](#)

- 12 **Klinzing, Niethard & Born. Systems memory consolidation during sleep**
Nature Neuroscience, 2019. PMID 31451802 [Open source](#)

- 13 **Wilson & McNaughton. Reactivation of hippocampal ensemble memories during sleep**
Science, 1994. DOI 10.1126/science.8036517 [Open source](#)

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| 14 | Rudoy et al. Strengthening individual memories by reactivating them during sleep
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| 15 | Girardeau et al. Selective suppression of hippocampal ripples impairs spatial memory
Nature Neuroscience, 2009. DOI 10.1038/nn.2384 Open source |
| 16 | Ellenbogen et al. Human relational memory requires time and sleep
PNAS, 2007. DOI 10.1073/pnas.0700094104 Open source |
| 17 | Langille. Remembering to Forget: A Dual Role for Sleep Oscillations
Frontiers in Cellular Neuroscience, 2019. PMID 30930746 Open source |
| 18 | de Vivo et al. Ultrastructural evidence for synaptic scaling across wake and sleep
Science, 2017. DOI 10.1126/science.aah5982 Open source |
| 19 | Diering et al. Homer1a drives homeostatic scaling-down during sleep
Science, 2017. DOI 10.1126/science.aai8355 Open source |
| 20 | Norimoto et al. Hippocampal ripples down-regulate synapses
Science, 2018. DOI 10.1126/science.aao0702 Open source |
| 21 | Li et al. REM sleep selectively prunes and maintains new synapses
Nature Neuroscience, 2017. DOI 10.1038/nn.4479 Open source |
| 22 | Yang et al. Sleep promotes branch-specific formation of dendritic spines
Science, 2014. DOI 10.1126/science.1249098 Open source |
| 23 | Nader & Hardt. A single standard for memory: the case for reconsolidation
Nature Reviews Neuroscience, 2009. DOI 10.1038/nrn2590 Open source |
| 24 | Memory Reconsolidation
NCBI Bookshelf Open source |
| 25 | Schultz. Dopamine reward prediction error coding
Current Opinion in Neurobiology, 2016. PMID 27069377 Open source |
| 26 | Shin et al. Continual Learning with Deep Generative Replay
NeurIPS, 2017. Adjacent engineering influence Open source |

A clear way to explain Cortex

A concise description that communicates the inspiration without overstating the science.

30-SECOND EXPLANATION

Cortex is an adaptive memory layer for AI agents. It borrows design principles from cognitive psychology and neuroscience: keep working memory small, organize long-term memory by type, reinforce information that actually helps, connect related experiences, consolidate knowledge offline, revise beliefs when evidence changes, and gradually archive stale material. It does not simulate a brain. It turns those principles into measurable software mechanisms intended to reduce irrelevant context, tool retries, latency, and token use while improving useful recall.

SAFE TO SAY	AVOID SAYING
Psychologically inspired memory architecture	Cortex works exactly like the human brain
Offline replay is analogous to consolidation	The agent literally sleeps or dreams
Outcome signals adapt retrieval utility	Cortex simulates dopamine
Cooling and archiving are reversible forgetting mechanisms	Biological pruning proves this algorithm is optimal
Benchmarks test whether the design improves agents	Neuroscience alone proves Cortex is faster

The one-line version

Cortex gives an agent a small working memory, an adaptive long-term memory, and an offline maintenance cycle - all evaluated as software, not presented as a synthetic brain.

CORTEX MEMORY

Open architecture. Inspectable memory. Measurable claims.